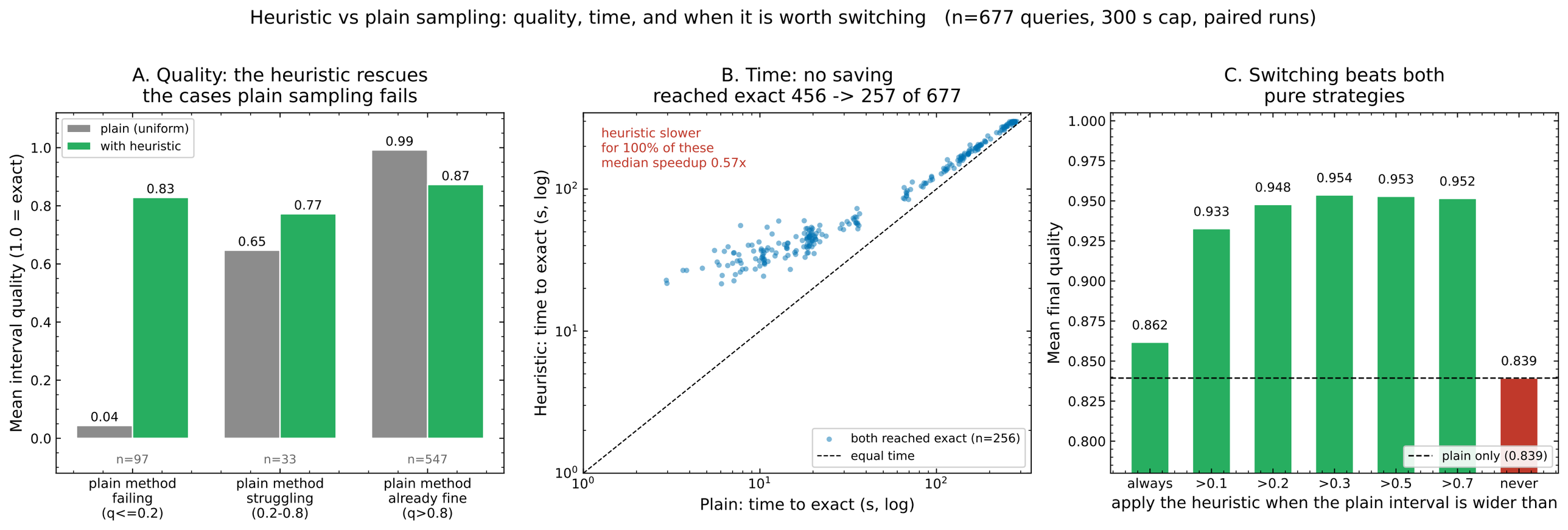
Heuristic vs plain sampling: quality, time, and when it is worth switching (n=677 queries, 300 s cap, paired runs)
A. Quality: the heuristic rescues the cases plain sampling fails
plain (uniform)
with heuristic
Mean interval quality (1.0 = exact)
0.04
0.83
0.65
0.77
0.99
0.87
n=97
n=33
n=547
plain method failing (q<=0.2)
plain method struggling (0.2-0.8)
plain method already fine (q>0.8)
B. Time: no saving reached exact 456 -> 257 of 677
heuristic slower for 100% of these
median speedup 0.57x
Heuristic: time to exact (s, log)
Plain: time to exact (s, log)
both reached exact (n=256)
equal time
C. Switching beats both pure strategies
Mean final quality
0.862
0.933
0.948
0.954
0.953
0.952
0.839
plain only (0.839)
always
>0.1
>0.2
>0.3
>0.5
>0.7
never
apply the heuristic when the plain interval is wider than